R331 · FINDING

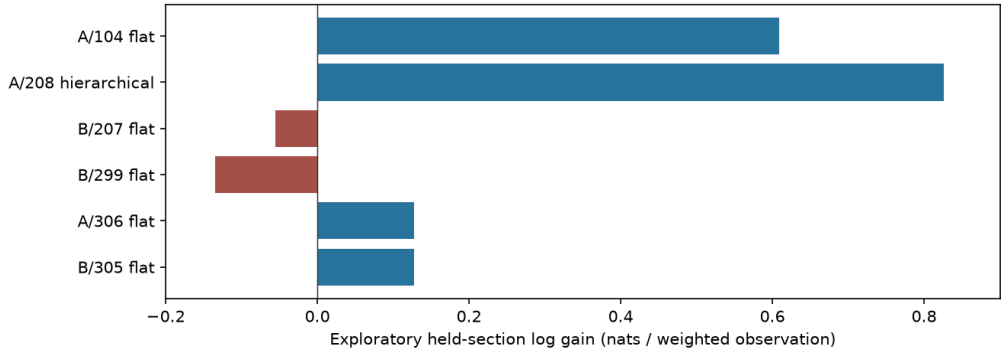
A useful 208 prediction is not the recovered design

The maker says the shortest repeat is about 200–399 and is not divisible by 13.

A: 2704 = 13 × 208. Its only allowed-range divisors, 208/338, violate that constraint.

B: 2691 = 13 × 207 = 9 × 299. Those lengths satisfy the maker constraint but predict poorly on this map.

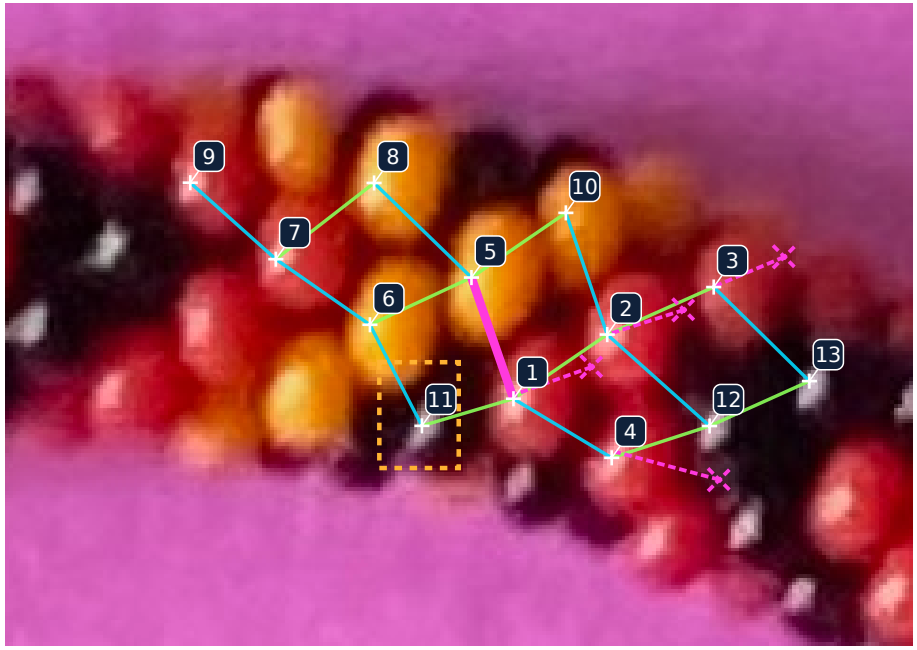
None of the saved maps establishes a design-compatible exact repeat. The 104/208 structure survives phase-preserving controls.



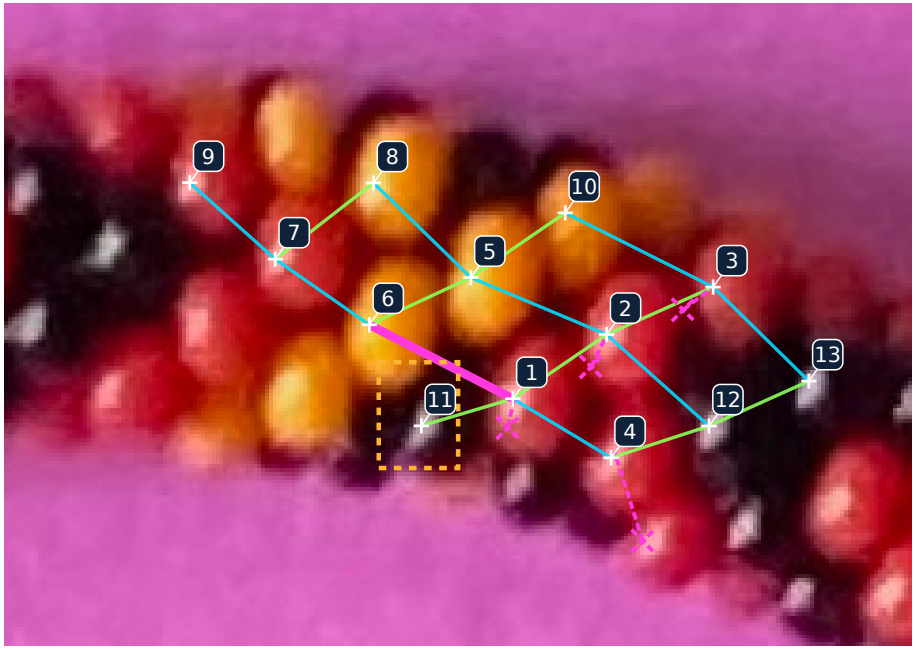
Same 900 original observations; all results are exploratory reuse. Hierarchical A/208 has an explicitly different prior; flat comparisons are in the report.

Four review problems follow. They ask for concrete observations in the photograph; they do not ask you to certify a model, count or pattern.

G: which row reaches the first red sequence?



working: magenta crosses predict red targets 1–4 from the same 11 colored references. Misses: 21.4, 21.2, 20.3, 28.6 px. Dashed amber box: dark material near support 11.



early: magenta crosses predict red targets 1–4 from the same 11 colored references. Misses: 9.1, 9.5, 10.0, 24.8 px. Dashed amber box: dark material near support 11.

Help I would use: compare the highlighted incoming link to red support 1 against the annotated yellow rows 6→5→10 and 7→6. Does the visible red-body sequence 1→2→3 continue that same diagonal, or bend into the adjacent row?

The local colored-target fit favors the alternative, but neither B/207 nor the weak 305/306 candidates resolves its downstream consequences. A/208 remains predictive evidence, not a design veto.

Cyan: +a, A+7/B+6. Green: +b, A+6/B+7. Magenta: disputed connection or colored-target prediction. White crosses mark recorded supports, **not measured bead centers**. Amber dashed rectangles mark visible-material inspection areas; they are not measured boundaries. Overlays can be hidden to inspect the literal original crop.

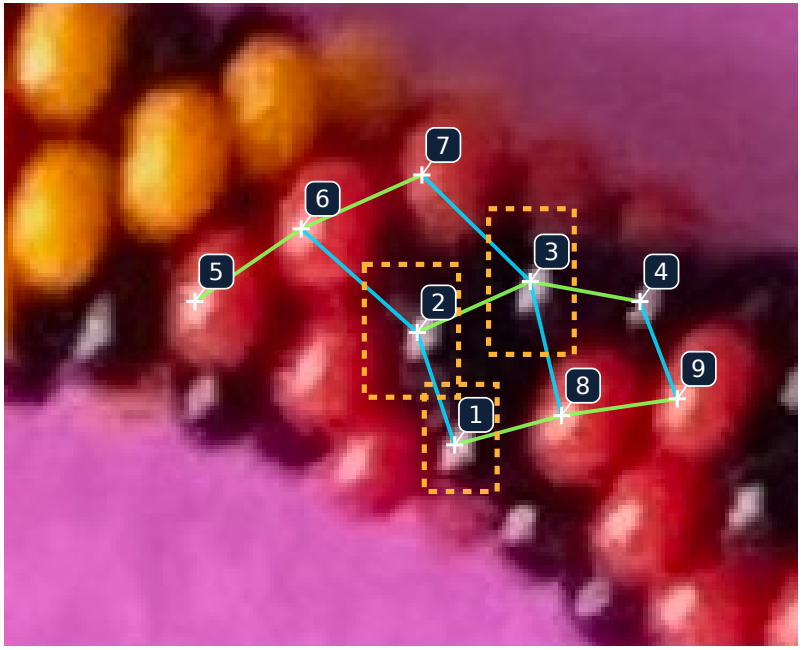
G: the exact competing assignments

# / original observation ID	Recorded color	working: (a,b); A / B	early: (a,b); A / B
1: P5-1645-0309	red	(8, 8); 104 / 104	(8, 7); 98 / 97
2: C-1669-0293	red	(8, 9); 110 / 111	(8, 8); 104 / 104
3: NEW-1697-280	red	(8, 10); 116 / 118	(8, 9); 110 / 111
4: E5-1671-0325	red	(9, 8); 111 / 110	(9, 7); 105 / 103
5: E5-1633-0278	yellow	(7, 8); 97 / 98	(7, 8); 97 / 98
6: E5-1607-0290	yellow	(7, 7); 91 / 91	(7, 7); 91 / 91
7: C-1582-0272	red	(6, 7); 84 / 85	(6, 7); 84 / 85
8: C-1608-0253	yellow	(6, 8); 90 / 92	(6, 8); 90 / 92
9: C-1559-0252	red	(5, 7); 77 / 79	(5, 7); 77 / 79
10: C-1659-0261	yellow	(7, 9); 103 / 105	(7, 9); 103 / 105
11: NEW-1621-315	black	(8, 7); 98 / 97	(8, 6); 92 / 90
12: K-1695-0317	black	(9, 9); 117 / 117	(9, 8); 111 / 110
13: K-1721-0308	black	(9, 10); 123 / 124	(9, 9); 117 / 117

The first three red target misses fall from 21.4/21.2/20.3 px to 9.1/9.5/10.0 px under the same 11-reference quadratic fit. These are partial-pigment extrapolations. The fourth target and wider reference fits remain sensitive. A concrete row identification here would justify a bounded replay; motif preference alone would not.

R331 · PROBLEM 2 OF 4

Black material: color label or wrong observation?



Same-direction comparison: green [0,+1] links dark supports 2→3→4 and red supports 5→6→7. The separate green link 1→8→9 shows the red-labeled site in its neighboring row. Amber boxes mark visible dark-material inspection areas.

Support 1, NEW-1706-342, was labeled red. Its appearance controls and A/208 favor black. Its B/207 prediction favors red.

Help I would use: inspect the dark material in box 1, then compare its visible body with boxes 2/3 and the adjacent red row in the same green direction. A reflection supplies an observation location; it does not measure the body center or establish pigment.

All six R330 red→black hypotheses remain provisional. More pattern agreement is not new confirmation.

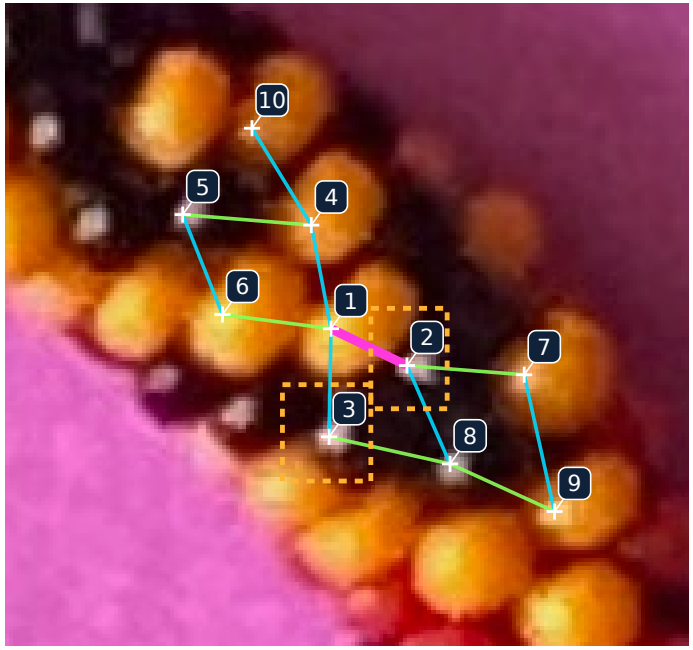
Cyan: +a, A+7/B+6. Green: +b, A+6/B+7. Magenta: disputed connection or colored-target prediction. White crosses mark recorded supports, **not measured bead centers**. Amber dashed rectangles mark visible-material inspection areas; they are not measured boundaries. Overlays can be hidden to inspect the literal original crop.

Black comparison: IDs, indices and competing color readings

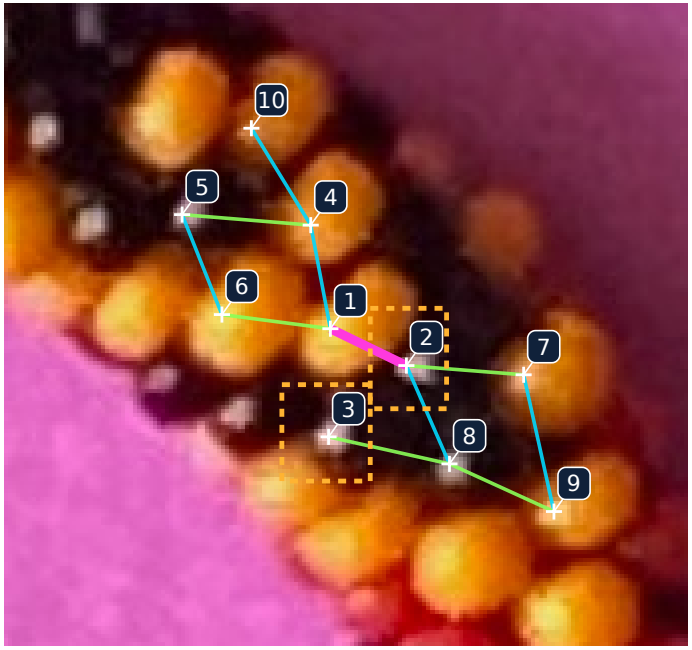
# / original observation ID	Recorded color	working: (a,b); A / B	Color hypothesis
1: NEW-1706-342	red	(10, 9); 124 / 123	red → black, provisional
2: K-1695-0317	black	(9, 9); 117 / 117	unchanged
3: K-1721-0308	black	(9, 10); 123 / 124	unchanged
4: D40-K-1746-0312	black	(9, 11); 129 / 131	unchanged
5: P5-1645-0309	red	(8, 8); 104 / 104	unchanged
6: C-1669-0293	red	(8, 9); 110 / 111	unchanged
7: NEW-1697-280	red	(8, 10); 116 / 118	unchanged
8: C-1729-0337	red	(10, 10); 130 / 130	unchanged
9: E5-1757-0332	red	(10, 11); 136 / 137	unchanged

The same image can support a dark-body reading while carrying a tinted bright patch. The review marks inspection areas rather than inferring complete black outlines. The full review includes matched contexts for all six hypotheses and three additional model flags; none changes the frozen labels.

Inward bend: one diagonal changes the conditional count



working: magenta joins supports 1→2. Working delta (0,−1); A−6/B−7. Amber boxes mark dark-material inspection areas. Nearby yellow supports carry the same cyan/green direction labels.



closest: magenta joins supports 1→2. Alternative delta (−1,0); A−7/B−6. Amber boxes mark dark-material inspection areas. Nearby yellow supports carry the same cyan/green direction labels.

Help I would use: compare 1→2 with the green/cyan yellow continuations and the dark material in boxes 2/3. Which family is supported by the visible body arrangement? The current native prediction favors working; the competing family survives complete replay.

Working H=(214,201), A/B counts 2704/2691. Closest alternative H=(215,200), counts 2705/2690. Replacing windows 124–126 is sufficient to produce that change.

Cyan: +a, A+7/B+6. Green: +b, A+6/B+7. Magenta: disputed connection or colored-target prediction. White crosses mark recorded supports, **not measured bead centers**. Amber dashed rectangles mark visible-material inspection areas; they are not measured boundaries. Overlays can be hidden to inspect the literal original crop.

Inward bend: exact labels and evidence

# / original observation ID	Recorded color	working: (a,b); A / B	closest: (a,b); A / B
1: C-1721-1566	yellow	(137, 131); 1745 / 1739	(138, 130); 1746 / 1738
2: K-1740-1575	black	(137, 130); 1739 / 1732	(137, 130); 1739 / 1732
3: MAT25-K-1719-1592	black	(136, 131); 1738 / 1733	(136, 131); 1738 / 1733
4: C-1717-1540	yellow	(138, 131); 1752 / 1745	(139, 130); 1753 / 1744
5: K-1683-1538	black	(138, 132); 1758 / 1752	(139, 131); 1759 / 1751
6: E5-1694-1563	yellow	(137, 132); 1751 / 1746	(138, 131); 1752 / 1745
7: C-1770-1577	yellow	(137, 129); 1733 / 1725	(137, 129); 1733 / 1725
8: K-1750-1601	black	(136, 130); 1732 / 1726	(136, 130); 1732 / 1726
9: C-1777-1612	yellow	(136, 129); 1726 / 1719	(136, 129); 1726 / 1719
10: C-1705-1514	yellow	(139, 131); 1759 / 1751	(140, 130); 1760 / 1750

Cached target-excluded errors, working → alternative: C-1721 6.11→9.56 px; K-1740 5.38→8.51 px; MAT25-K-1719 5.22→12.59 px. Native-response ratios for K-1740: .691→.057. These measure response at predicted supports, not distances between measured body centers. Better local body correspondence would matter more than another pattern score.

F: a common image shift still leaves material unexplained



frozen: magenta is the cached visible model-body boundary; orange marks measured yellow/red pixels outside their assigned model body. No raster translation.



translated: magenta is the cached visible model-body boundary; orange marks measured yellow/red pixels outside their assigned model body. Raster moved +8 px right / 4 px up; fitted only at yellow support 1.

Help I would use: compare the orange measured pigment at targets 2/3 with the magenta assigned-body borders after translation. Identify any visibly wrong body association before changing shape or lighting. A visual association can narrow correspondence; the crop alone cannot choose the physical parameter responsible.

Reference target 1 improves from about 59% to 99% coverage. Targets 2/3 remain near 49%. The frozen 3000-body model is incompatible with the photographic conditional count; this is a retained diagnostic, not a reconstruction proposal.

Cyan: +a, A+7/B+6. Green: +b, A+6/B+7. Magenta: disputed connection or colored-target prediction. White crosses mark recorded supports, **not measured bead centers**. Amber dashed rectangles mark visible-material inspection areas; they are not measured boundaries. Overlays can be hidden to inspect the literal original crop.

F: what the crop can and cannot decide

# / original observation ID	Recorded color	working: (a,b); A / B
1: C-1302-0316	yellow	(0, 0); 0 / 0
2: C-1391-0312	red	(2, 2); 26 / 26
3: C-1422-0264	red	(2, 4); 38 / 40
4: C-1251-0290	yellow	(-2, 0); -14 / -12
5: C-1277-0302	yellow	(-1, 0); -7 / -6
6: C-1339-0277	red	(0, 2); 12 / 14
7: P5-1364-0299	red	(1, 2); 19 / 20
8: NEW-1442-248	red	(2, 5); 44 / 47

Targets 1–3 use frozen model addresses −64/−38/−26. Orange pixels are actual measured colored regions; magenta borders come from the saved visibility raster. The source image and all 3000 model bodies are unchanged.

A visible mistake in which body owns a colored region would redirect correspondence work. If the association is right, the residual belongs to projection/shape/trajectory diagnosis. No new fitting campaign was performed in R331.

R331 · NEXT INVESTIGATION

Resolve one row continuation before another global fit

Most valuable next investigation: a tightly bounded, color-independent body-correspondence audit at G, using the same visible diagonal in neighboring colored and black sequences, followed by replay of only the already implicated windows.

The new latent model separates color substitutions from shift proposals and abstains when several ±1/±6/±7 corrections explain the same color. It flags 52/144 injected isolated A-index errors but uniquely restores none from the motif alone. Frozen neighbor equations restore 138/144 bookkeeping errors; they cannot validate an internally consistent wrong row.

Unknown motif entries remain unknown. No winding, count, exact design or provisional color correction is adopted. Stop for review; no automatic R332.

Report: photo2/R331_REPORT.md. Full offline review: index.html. Reproduction and machine-readable evidence are linked there. These four review tasks do not establish new photographic truth until a specific visible choice is recorded.